发机前准备工作指南

Pre-departure Preparation Guide

1. 请按购买设备功率段准备纯净水，具体的要求如下：

Please prepare pure water at the power section of the purchased equipment as follows:

机型Model	功 率 Power	容 积 Volume
不分机型 No model	≤2000W	20L
不分机型 No model	3000W	55L
不分机型 No model	4000W	70L
不分机型 No model	6000W	100L
不分机型 No model	8000W	110L
不分机型 No model	10000W	170L
不分机型 No model	12000W	200L

注：冷水机主要是用于冷却切割头和激光器重要部件，使用冷却水要求必须是纯净水或蒸馏水，导电率小于10uS/cm（推荐采用屈臣氏蒸馏水）。

Note: The chiller is mainly used for cooling cutting head and laser important parts, the cooling water must be pure water or distilled water, recommended for conductivity less than 10uS/cm().

2. 请准备外接电源线及电源开关，具体要求如下：

Please prepare external power cord and power switch, as follows:

要求：三相五线制（火、火、火、零、地）380V50Hz，三相不平衡度<2.5%，电压波动<5%，单相 220V±5。

Requirements: Three-phase five-line system (fire, fire, fire, zero, ground) 380V50Hz, three-phase unbalance <2.5%, voltage fluctuation <5%, single phase 220V ± 5.

地线要求：接地电阻不应大于 4 欧姆，地线接地深度不能低于 2m。

Requirements: Three-phase five-line system (fire, fire, fire, zero, ground) 380V50Hz, three-phase unbalance
<2.5%, voltage fluctuation <5%, single phase 220V ± 5.

电源开关、电源线要求：

Power switch and power cord requirements:

功率 Power	电源开关 Power switch	电源线 Power core	电源线材质 Power core material
1500W 以下 <=1500w	总电源：三相 60A 空气开关（带漏电保护） 抽风机：三相 32A 空气开关（不带漏电保护） 如标配空压机：三相 60A 空气开关（不带漏电保护） 冷干机：单相 32A 空气开关（带漏电保护） Main power supply: three-phase 60A air switch (with electric leakage protection) Extraction fan: three-phase 32A air switch (without electric leakage protection)	6平方线 6-Square Line	铜 copper

	If standard air compressor: three-phase 60A air switch (without leakage protection) Cold dryer: single-phase 32A air switch (with electric leakage protection)		
3000W 以下 **<=300w**	总电源：三相 60A 空气开关（带漏电保护） 抽风机：三相 60A 空气开关（不带漏电保护） 如标配空压机：三相 60A 空气开关（不带漏电保护） 冷干机：单相 32A 空气开关（带漏电保护） Main power supply: three-phase 60A air switch (with electric leakage protection) Extraction fan: three-phase 60A air switch (without electric leakage protection) If standard air compressor: three-phase 60A air switch (without leakage protection) Cold dryer: single-phase 32A air switch (with electric leakage protection)	10平方线 10-Square Line	铜 copper
6000W 以下 **<=6000w**	总电源：三相 100A 空气开关（带漏电保护） 抽风机：三相 60A 空气开关（带漏电保护） 如标配空压机：三相 60A 空气开关（不带漏电保护） 冷干机：单相 32A 空气开关（带漏电保护） Main power supply: three-phase 100A air switch (with electric leakage protection) Extraction fan: three-phase 60A air switch (with electric leakage protection If standard air compressor: three-phase 60A air switch (without leakage protection) Cold dryer: single-phase 32A air switch (with electric leakage protection)	16平方线 16-Square Line	铜 copper
8000-12000W	总电源：三相 100A 空气开关（带漏电保护） 抽风机：三相 60A 空气开关（带漏电保护） 激光器： 冷水机： 如标配空压机：三相 60A 空气开关（不带漏电保护） 冷干机：单相 32A 空气开关（带漏电保护） Main power supply: three-phase 100A air switch (with electric leakage protection) Extraction fan: three-phase 60A air switch (with electric leakage protection) Laser unit: Chiller: If standard air compressor: three-phase 60A air switch (without leakage protection) Cold dryer: single-phase 32A air switch (with electric leakage protection)	16平方线 16-Square Line	铜 copper

3 . 稳压器具体要求如下：

Specific requirements for the voltage stabilizer are as follows:

功率 Power	稳压器 Voltage stabilizer	备注 Remark
1500W	30KVa	
3000w	50KVa	
4000-6000w	80KVa	
8000-12000w	120KVa	

4 . 切割辅助气体和减压阀具体要求如下：

The specific requirements for cutting auxiliary gas and pressure relief valves are as follows:

气体种类 Gas type	气压 Pressure	气体纯度 Gas purity	气化器要求 Vaporizer requirement
氧气 Oxygen	1MPa	99.9%	
氮气 Nitrogen	2MPa	99.99%	

注：（如图1.2.3 所示）

Remark: (As shown in Picture 1,Picture2 and Picture3)

使用液氧必须要配备汽化器大于或等于 30 立方每小时

Using liquid oxygen must be equipped with vaporizer greater than or equal to 30 cubic meters per hour

使用液氮必须要配备气化器大于或等于 50 立方每小时

Using liquid nitrogen must be equipped with vaporizer greater than or equal to 50 cubic meters per hour

6000w 以上机器必须要配备气化器大于或等于 100 立方每小时

Machine above 6000w must be equipped with vaporizers greater than or equal to 100 cubic meters per hour.



图1 液态气体

Picture1 liquid gas



图2 汽化器

Picture2 vaporizer



图3 汽化器参数

Picture3 vaporizer parameter

5. 减压阀 （如图 4、5 所示）

Pressure relief valve (as shown in Picture 4 and Picture 5)

氧气减压阀： 要求输出压力 0.1-1.5MPa

（推荐：美国捷锐 652x-175）

Oxygen pressure relief valve: required output pressure 0.1-1.5MPa, (recommended: US Jelite 652x-175).

氮气减压阀：要求压力 1-2.5MPa

（推荐：使用美国捷锐 452IN-450）

Nitrogen pressure relief valve: required output pressure 1-2.5MPa, (Recommended: US Jelite 452IN-450).



图4 捷锐氮气减压阀

Picture4 US Jelite 452IN-450 for Nitrogen pressure relief valve



图5.捷锐氧气减压阀

Picture5 US Jelite 652x-175 Oxygen pressure relief valve

空气（空压机参数要求）

Air (air compressor parameter requirements)

			Laser power激光器功率			
	名称 Description	2000W以下 <2000w	3000-4000W	6000-8000W	10000-12000W	备注 Remark
1	空压机排气量 Air discharge (m3/min)	1.0	1.2	2.0	2.0	

	排气压力 Air discharge pressure (Bar)	16	20	20	25	
2	冷冻式干燥机处理气量 Cold dryer air handle(m3/min)	2.0	2.5	3.8	3.8	
	工作压力 Working pressure(Bar)	16	20	30	30	
3	预过滤器：除水除油 Pre-filter: remove water and oil					
	空气含水量 Water content(um)	0.3	0.3	0.3	0.3	
	空气含油量 Oil content(ppm)	0.3	0.3	0.3	0.3	
4	前置精密过滤器：除水 Pre-precision filter:remove water					
	空气含水量 Water content(um)	0.01	0.01	0.01	0.01	
	空气含油量 Oil content(ppm)	0.01	0.01	0.01	0.01	
5	后置精密过滤器：除油 Post-precision filter:remove oil					
	空气含水量 Water content(um)	0.01	0.01	0.01	0.01	
	空气含油量 Oil content(ppm)	0.003	0.003	0.003	0.003	

6	后置精密过滤器：除微油雾及颗粒 Post-precision filter:remove micro-oil fog and particle					
	空气含水量 Water content(um)	0.01	0.01	0.01	0.01	
	空气含油量 Oil content(ppm)	0.001	0.001	0.001	0.001	
7	储气罐 Gas tank					
	容积 Volume(m3)	0.6	0.6	1.0	1.0	
	工作压力 Working pressure(Mpa)	3.3	3.3	3.3	3.3	
8	吸附式干燥机（自带精密过滤器） Adsorbent dryer (with its own precision filter)	不需要使用 Not necessary	必须配备 Must equipped	必须配备 Must equipped	必须配备 Must equipped	
	压力 Pressure(Mpa)		2.0	2.0	2.5	
	处理气量 Air handle(m3/min)		2	2.8	2.8	

6. 客户根据不同激光器功率准备不同厚度板材，具体要求如下

Customer shall prepare different thickness plates according to different laser power, and the specific requirements are as follows

功率 Power	铁板 Carbon steel	不锈钢 Stainless steel	铝 Aluminum	黄铜 Brass

500W	1-5mm	1-3mm		
750/800W	1-8mm	1-3.5mm	1mm 以下	1mm 以下
1000w	1-10mm	1-4mm	1-2mm	1-1.5mm
1500w	1-12mm	1-5mm	1-3mm	1-2mm
2000w	1-16mm	1-8mm	1-5mm	1-3mm
3000w	1-20mm	1-10mm	1-8mm	1-5mm
4000w	1-22mm	1-12mm	1-10mm	1-8mm
6000w	1-22mm	1-14mm	1-12mm	1-10mm
8000w	1-25mm	1-16mm	1-14mm	1-14mm
10000w	1-30mm	1-20mm	1-16mm	1-16mm
12000w	1-35mm	1-22mm	1-18mm	1-18mm

7. 地脚板根据不同机型准备地脚板数量

Number of fooboards prepared according to different models

主要目的：让设备在高速运动不影响机械性能和切割板材晃动，设备地脚图请看详细附图。

Main purpose: to move the equipment at high speed does not affect the mechanical performance and cutting plate shaking, please see the detailed drawings.

8. 设备环境

Facility environment

要求通风条件好、无腐蚀、无漏水的场地环境，尽量避免靠近受电磁波干扰设备附近，如氩弧焊、电焊。

Site environment with good ventilation conditions, no corrosion and no water leakage, and avoid being near the electromagnetic wave interference equipment, such as argon arc welding and electric welding.

9. 地面地基要求

Ground foundation requirements

1. 水泥地方面层不能小于或等于 150mm
Cement floor aspect layer shall not be less than or equal to 150mm

2. 地面必须要平整，平整度正负 10mm
Ground must be flat, positive or negative 10mm

3. 摆放设备周围 10m 内不能有产生振动设备，如有产生振动设备必须要挖防震沟，防震沟挖断水泥层还需要再挖深 10cm.

There cannot be vibration equipment within 10m around the placing equipment. If there is vibration equipment, dig the seismic trench and dig the cement layer by 10cm.

10. 激光器空调房 （如图 6.7 所示）

Laser air-conditioning room (as shown in Picture 6 ,Picture 7)

3000w 以上设备必须要配备空调房，空调房尺寸大约 5*2.5 米或 3.5*3.5 米

Equipment above 3000w(multimode) must be equipped with air conditioning room .The room dimension is about 5*2.5m or 3.5*3.5m

Picture6



Picture7



11. 清洁使用耗材 （图 8.9 所示）

Clean consumable(as shown in Picture 8,Picture9)

1. 无水乙醇≥99.7%一瓶500 毫升或异丙醇一瓶
Anhydrous Ethanol ≥ 99.7% (500 ml/ bottle) or a bottle of isopropyl alcohol.

2. 美文纸和透明胶各 2 卷
Textured paper tape and transparent film 2 rolls per each.





图8 异丙醇
Picture8 isopropyl alcohol

图9 美纹纸
Picture9Textured paper tape

12. 根据不同型号设备卸车需要请叉车或者吊车具体要求如下：

Specific requirements for forklift or crane according to different models of equipment are as follows

F3015E F3015BE F4015E F4015BE 3015JE F3015KE

请准备 **5** 吨叉车和两台手动 **3** 吨液压叉车

Please prepare a 5 ton forklift and two manual 3 ton hydraulic forklifts

F3015HE F3015HBE F3015HDE F3015JDE F4020E F4020BE F4015JDE F6015JE F6015E F6010GE F6015GE

请准备 **8** 吨叉车和两台手动 **5** 吨液压叉车

Please prepare an 8 ton forklift and two manual 5 ton hydraulic forklifts

F3015HDE F3015HBDE F4020HE F4020HBE F4020HDE F6015BE F6020E F6020BE F6025E F6020JE F6020GE F6035GE

请准备 **10** 吨叉车和两台 **5** 吨液压叉车

Please prepare a 10 ton forklift and two 5 ton hydraulic forklifts

F4020HDE F4020HBDE 6320JDE

请准备 10 吨叉车或 20 吨吊车

和两台 5 吨手动液压叉车

Please prepare a 10 ton forklift or a 20 ton crane and two 5-ton manual hydraulic forklifts

F6015HDE F6015HBDE F6020HE F6020HBE F6020HDE F6020HBDE F8025E TC622A3 F1035GE F1238GE

请准备 20 吨吊车和 10 吨手动液压叉车

Please prepare a 20 ton crane and a 10 ton manual hydraulic forklift

F6025HDE F6325TDE F1225E

请准备1台30吨吊车和2台8 吨叉车

Please prepare 130 ton crane and 28 ton forklift trucks

F8025TDE

请准备1台40吨吊车和 2台8 吨叉车

Please prepare 140 ton crane and 28 ton forklift trucks

F1225TE F1225TDE

请准备1台50吨吊车和 2台10 吨叉车

Please prepare 150 ton crane and 210 ton forklift trucks

13. 受训人员 Trainee

1. 人数1-2 名
   one or two trainee

2. 平面机型4000W以下

   Laser power < =4000W for model KE\E\HE\HDE.

要求：懂基本电脑使用，懂基本CAD画图，随机应变能力强。学历中专，年龄不要超过40岁，然后懂机械原理。

Requirements: understand the basic computer use, understand the basic CAD drawing, random strain ability. Technical secondary school, the age of

operatore is not more than 40 years old, and then understand the mechanical principles.

3. 管板一体机型4000W以下

Laser power < =4000W for model BE\HBE\HBDE.

要求：懂基本电脑使用，懂基本CAD画图，懂得三维软件solidworks,随机应变能力强。学历中专，年龄不要超过40岁，然后懂机械原理。

Requirements: understand the basic computer use, understand the basic CAD drawing, understand the 3 D software solidworks, random strain ability. Technical secondary school, the age is not more than 40 years old, and then understand the mechanical principles.

4. 平面机型6000W以上

Laser power = >6000W for model KE\E\HE\HDE.

要求：懂基本电脑使用，懂基本CAD画图，随机应变能力强。有一年以上3000W以上开机工作经验。学历中专，年龄不要超过40岁，然后懂机械原理。

Requirements: understand the basic computer use, understand the basic CAD drawing, random strain ability. More than one year of 3000W boot up experience. Technical secondary school, the age is not more than 40 years old, and then understand the mechanical principles.

5. 管板一体机型6000W以上

Laser power = >6000W for model BE\HBE\HBDE.

要求：懂基本电脑使用，懂基本CAD画图，懂得三维软件solidworks, 随机应变能力强。有一年以上3000W以上开机工作经验。学历中专，

年龄不要超过40岁，然后懂机械原理。

Requirements: understand the basic computer use, understand the basic CAD drawing, understand the 3 D software solidworks, random strain ability. More than one year of 3000W boot up experience. Technical secondary school, not older than 40, and then understand mechanical principles.

6. 专业切管机 Professional pipe cutting machine

要求：懂基本电脑使用，懂得三维软件solidworks, 随机应变能力强。学历中专，年龄不要超过40岁，然后懂机械原理。

Requirements: understand the basic computer use, understand the 3 D software solidworks, random strain ability. Technical secondary school, the age is not more than 40 years old, and then understand the mechanical principles.

Thanks for your cooperation!

We will serve you wholeheartedly!

Foshan Hui Parkson Laser Technology Co., Ltd